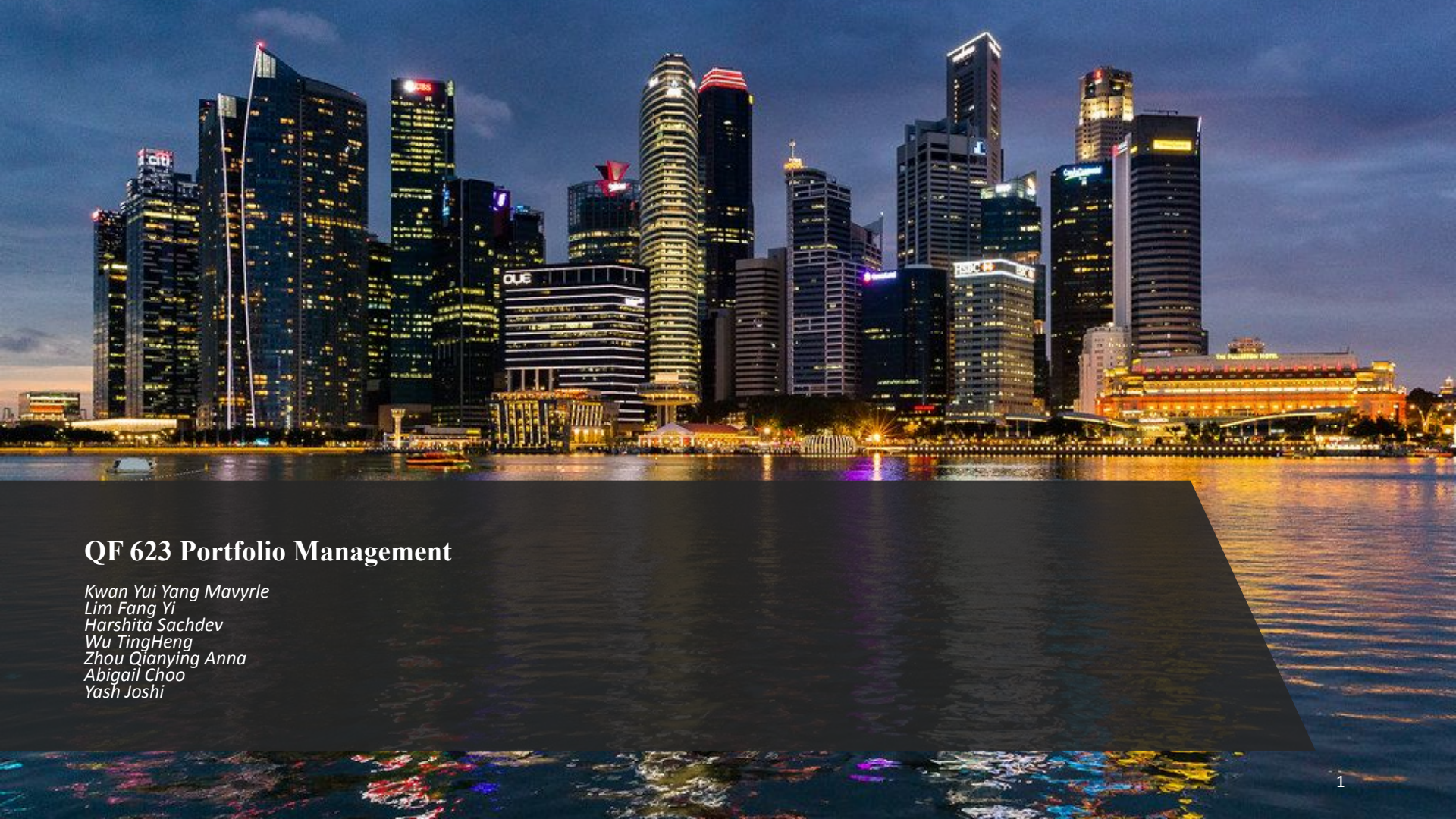




QF 623 Portfolio Management

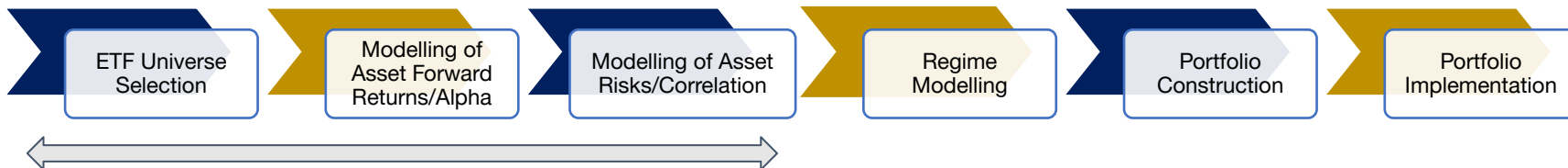
*Kwan Yui Yang Mavryle
Lim Fang Yi
Harshita Sachdev
Wu TingHeng
Zhou Qianying Anna
Abigail Choo
Yash Joshi*

Portfolio Management Overview

Objectives, investment strategy, and portfolio management framework

1. The objectives of the investment portfolio are to:
 - Achieve **consistent and strong risk-adjusted returns**
 - **Outperform the market index** in the long-run
 - **Mitigate steep market drawdowns** during “risk-off “ market environments
2. Investment objectives are achieved by:
 - **Diversification** across asset classes - allows **for tactical rotations** to uncorrelated assets; **reduce exposure** to **market risk premium** in times of sharp market downturns
 - Identification of key return drivers for broad asset classes which will drive returns/alpha forecasts - **enhance returns through exposure to positive idiosyncratic risk**
 - Economic regime modelling to **model probability of market shocks** - input for portfolio construction

3. Portfolio Management Framework:



ETF Investment Universe Selection

Objectives, investment strategy, and portfolio management framework

	Ticker	Asset Class	Inception	Expense Ratio	AUM (USD bn)	Regime Role	Rationale for inclusion
1	SPY	U.S. Large-Cap Equity	Jan 22 1993	0.09%	400	Equity Benchmark	Broad large-cap U.S. equity; ultra-liquid low-cost beta anchor.
2	MTUM	U.S. Equity Momentum	Apr 16 2013	0.15%	25	Momentum Tilt	Systematic U.S. price momentum factor; enhances trend following in risk-on regimes.
3	VLUE	U.S. Equity Value	Apr 16 2013	0.15%	10	Value Tilt	Large-cap value factor exposure; mean-reversion anchor in valuation sell-offs.
4	USMV	U.S. Equity Minimum Volatility	Oct 18 2011	0.15%	35	Defensive Tilt	Low-volatility U.S. equities; cushions drawdowns in high-vol regimes.
5	EFA	Developed ex-U.S. Equity	Aug 14 2001	0.32%	70	Global Equity Diversifier	Diversified developed-markets exposure; reduces home-bias in equity allocation.
6	AGG	U.S. Aggregate Bond	Sep 22 2003	0.04%	120	Risk-Off Anchor	Core USD investment-grade bonds; ballast in equity drawdowns and rising-rate regimes.
7	HYG	U.S. High-Yield Credit	Apr 04 2007	0.49%	30	Credit Carry	High-yield corporate carry; boosts income in stable-credit regimes.
8	TLT	U.S. Long-Duration Treasuries	Jul 22 2002	0.15%	15	Duration Tilt	Long-dated Treasury exposure; macro hedge when growth slows.
9	TIP	Treasury Inflation-Protected (TIPS)	Dec 04 2003	0.19%	25	Inflation Hedge	Real-return protection; key in high-inflation regimes.
10	GLD	Physical Gold Shares	Nov 18 2004	0.40%	50	Tail-Risk Diversifier	Gold as store-of-value and crisis hedge; diversifies equity/bond risk.
11	DBC	Broad Commodity Index Fund	Feb 03 2006	0.89%	6	Commodity Carry	Broad commodity roll-yield; positive carry in backwardation regimes.
12	VNQ	U.S. Real Estate (REITs)	Sep 23 2004	0.12%	40	Income-Generating Tilt	U.S. real estate income; diversifies equity beta with rental yield.
13	BIL	Ultra-Short U.S. T-Bills	May 25 2007	0.14%	20	Cash Proxy	Near-cash equivalent; safe-haven in volatility spikes and deflationary regimes.

Regime Modelling

MSCI rule-based 4 regimes benchmark: Intro

Objective: To classify the global economic environment into one of four distinct regimes based on signals for growth and inflation.

The Four Regimes:

1. Goldilocks: Strong Growth, Falling Inflation
2. Heating Up: Strong Growth, Rising Inflation
3. Stagflation: Slowing Growth, Rising Inflation
4. Slow Growth: Slowing Growth, Falling Inflation

Key Ingredients:

1. Growth Signal: Composite Leading Index from both China and United States.
2. Inflation Signal: CPI from United States.

Regime Modelling

MSCI rule-based 4 regimes benchmark: Formulas

Intuition:

1. To measure **momentum** or **acceleration** not just the direction
2. Calculates the chg. in YoY growth rate. “Is the economy’s growth **speeding up** or **slowing down** compared to a **few months ago**?”



MSCI ECONOMIC REGIME ALLOCATOR INDEXES METHODOLOGY | JANUARY 2025

3) Rules to Assign Economic Regimes

Economic Regime	Short-term and Long-term Signals
Stagflation	$(US - CLIYOY_{qoq\ change} \leq 0 \text{ and } CN - CLIYOY_{qoq\ change} \leq 0) \text{ and } (US - CPIYOY_{qoq\ change} > 0)$
Heating Up	$(US - CLIYOY_{qoq\ change} > 0 \text{ or } CN - CLIYOY_{qoq\ change} > 0) \text{ and } (US - CPIYOY_{qoq\ change} > 0)$
Slow Growth	$(US - CLIYOY_{qoq\ change} \leq 0 \text{ and } CN - CLIYOY_{qoq\ change} \leq 0) \text{ and } (US - CPIYOY_{qoq\ change} \leq 0)$
Goldilocks	$(US - CLIYOY_{qoq\ change} > 0 \text{ or } CN - CLIYOY_{qoq\ change} > 0) \text{ and } (US - CPIYOY_{qoq\ change} \leq 0)$

De-coding the Signals:

- **US-CLIYOY**qoq chg > 0:
US **growth** accelerating
- **US-CPIYOY**qoq chg > 0:
US **inflation** accelerating

$$US - CLIYOY_{qoq\ change} = \frac{US - CLI_{(t)}}{US - CLI_{(t-12m)}} - \frac{US - CLI_{(t-3m)}}{US - CLI_{(t-15m)}}$$
$$CN - CLIYOY_{qoq\ change} = \frac{CN - CLI_{(t)}}{CN - CLI_{(t-12m)}} - \frac{CN - CLI_{(t-3m)}}{CN - CLI_{(t-15m)}}$$
$$US - CPIYOY_{qoq\ change} = \frac{US - CPI_{(t)}}{US - CPI_{(t-12m)}} - \frac{US - CPI_{(t-3m)}}{US - CPI_{(t-15m)}}$$

Regime Modelling

Finding the next regime using SVM

Objective: To create stable and forward-looking regime prediction model that filters short-term “noise”

Features:

- Global growth:
Lagged US & CN CLI
- Inflation pressure:
Short-term vs long-term trend CPI
- Monetary policy stance:
FED funds

Walk-Forward approach:

- Train before 2016
- Expand rolling window 1 month basis
- 0 data leak like `train_test_split()`
- RBF to handle unknown data rs

Benchmark comparison:

- Low accuracy != bad thing
- Benchmark data is “noisy” due to frequent flickering regime chg.
- Low turnover & costs
- Deliver actionable signals

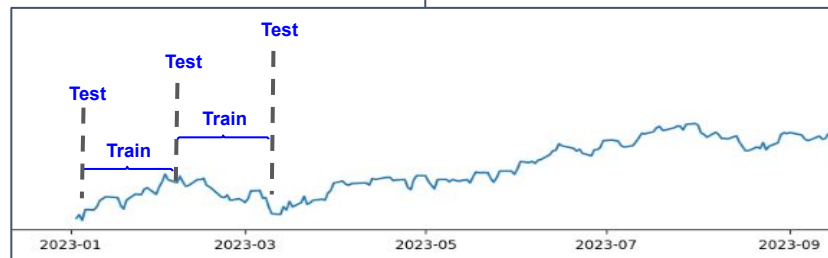
Regime Modelling

Finding the next regime using SVM



Walk-forward w expanding window training:

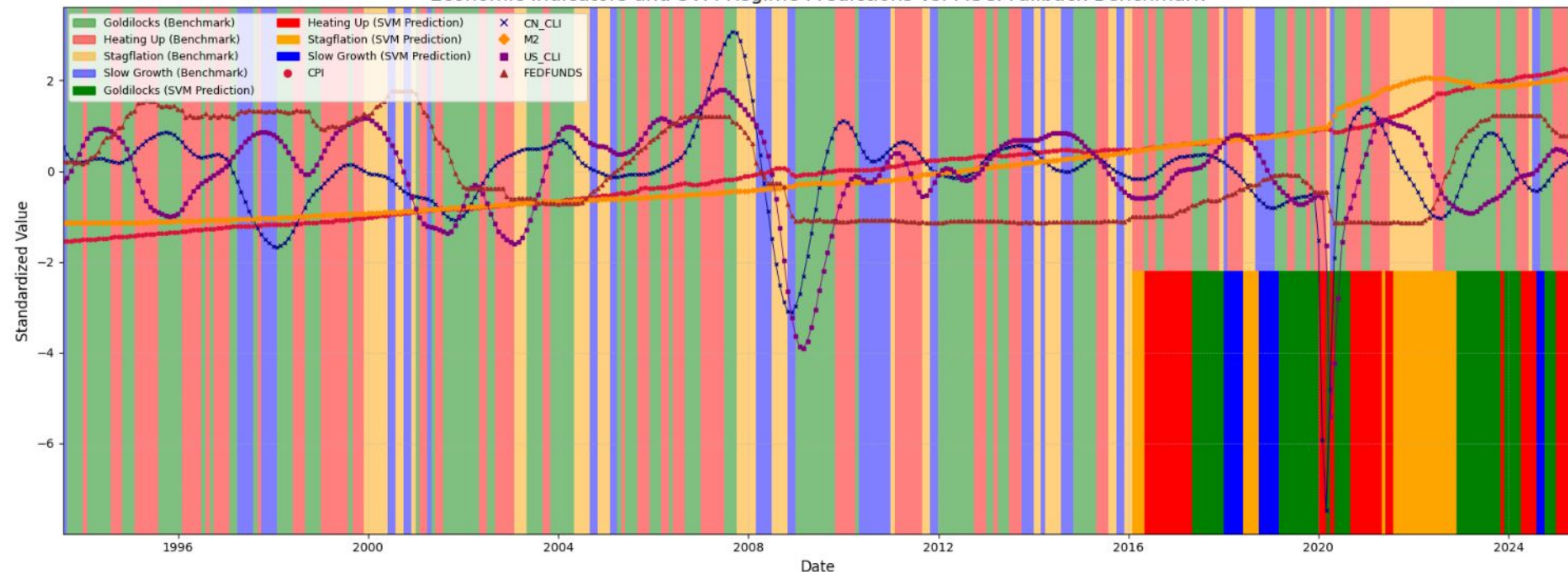
1. Walk-forward: Model gets “smarter” overtime
2. Rolling window: Rolling till the cutoff date



Regime Modelling

Benchmark vs Prediction

Economic Indicators and SVM Regime Predictions vs. MSCI Fallback Benchmark



--- SVM Performance Against MSCI Fallback Benchmark ---

Walk-forward overall accuracy: 0.634

1.0 0.633929

0.0 0.366071

Name: frequency, dtype: float64

	Pred_Stagflation	Pred_Heating Up	Pred_Slow Growth	Pred_Goilocks
True_Stagflation	13	2	0	2
True_Heating Up	7	23	4	11
True_Slow Growth	1	2	6	0
True_Goilocks	3	7	2	29

Overview

ETF Selection

Regime Model

Asset Returns Model

Asset Risks

Portfolio Construction

Attribution

Modelling of Asset Forward Returns/Alpha

Feature Engineering for Prediction



Risk Premia Factors

Excess Market Returns

Size (SMB)

Value (HML)

Profitability (RMW)

Investment Style (CMA)

Macroeconomic

Median CPI

Unemployment Rate

10-year Real Interest rate

Predicted Goldilocks Regime

Predicted Heating Up Regime

Predicted Stagflation Regime

Predicted Slow Growth Regime

Idiosyncratic

Rolling 6m Returns

1 year Z-Score

Rolling 6m Skew

Rolling 6m Realized Vol

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Modelling of Asset Forward Returns/Alpha

Alpha Framework

Alpha Type	Signal	Lookback	Alpha Role
1. Price Momentum	6-Month Momentum	126 trading days	Medium-term trend capture
	12-Month Momentum	252 trading days	Long-term trend anchor
2. Mean Reversion	5-Day Reversal	5 trading days	Short-term pullback exploitation
	20-Day Reversal	20 trading days	Medium-pullback signal
3. Trend & Vol Filters	MA200 Crossover	200 trading days	Entry/exit gate
	20-Day ATR Breakout	20 trading days	Volatility breakout trigger
	20-Day Realized Vol Filter	20 trading days	Dampens signals in high-vol regimes
4. Carry & Macro	Bond Carry (YTM - 3M T-Bill)	Spot	Income in stable/slow-growth
	Yield Curve Slope (10Y - 3M)	Spot	Macro regime timing
5. Style Tilts	Value (VLU): NAV/Price – 1	Spot	Valuation mean reversion
	Low Vol (USMV): $1/\sigma_{20}$	20 trading days	Defensive stability tilt
	Quality (QMJ) Factor	Daily	Quality income buffer
	Momentum (UMD) Factor	Daily	Cross-checked factor momentum
6. Liquidity & Sentiment	Volume Z-Score	20 trading days	Spike/liquidity signal
	Implied Vol Filter (VIX/VVIX)	Threshold	Risk-sentiment gate

Modelling of Asset Forward Returns/Alpha

Modeling Returns



Models

Considerations

Linear Regression

ElasticNet

XGBoost

Training

- Rolling window prevent lookahead bias
- Each ETF gets its own model with set of features (due to idiosyncratic, features)

- Linear Regression: sets as a benchmark
- Complex models do not necessarily beat simpler models
- Willing to give up a bit of accuracy for better speed, explainability of model
- Training cost for complex models, especially with need for rolling approach and scaling up number of assets

- ElasticNet shrinks coefficients and L1 penalty zeroes out factors that do not explain alphas adequately (feature selection)
- Regularization reduces overfit. L2 shrinkage reduces coefficients for correlated factors, addressing multicollinearity weakness
- Small weights also ensure out of sample stability -> more robust to regime shifts

- Ensemble of boosted decision trees learn non-linear relationships, threshold effects and factor interactions (e.g., ΔVIX matters only when momentum is negative)
- Gradient boosting re-weights difficult observations each round → adapts to varying noise levels across regimes., unlike LR which assume constant residual variance across states
- Robust: Loss function + tree structure isolate outliers to their own leaves

Modelling of Asset Risk: Covariance Matrix

Approaches to obtain covariance matrix

Choice of covariance estimator significantly affects the stability and robustness of the optimized portfolio weights

Stabilize optimized weights by shrinking sample covariance matrix

Sample Covariance

- Unbiased, but noisy and unstable covariance matrix becomes nearly-singular
- Risk overfitting: sensitive to minor changes in returns of each asset
- Non-robust, unstable weights which require portfolio manager to deploy extreme leverage

Ledoit-Wolf Shrinkage

- Works well when we need to do optimize weights on a rolling basis in the lookback window
- Especially when there is need to scale up number of assets, and insufficient observation of returns compared to number of assets
- Apply a shrinkage estimator with optimal shrinkage intensity that minimize MSE between true covariance matrix and estimate
- Bias is introduced to reduce overfitting

Oracle Approximating Shrinkage

- Improvement on Ledoit-Wolf assuming multivariate normality of returns.
- Uses a different formula for optimal shrinkage intensity, otherwise similar shrinkage structure

Modelling of Asset Risk: Covariance Matrix

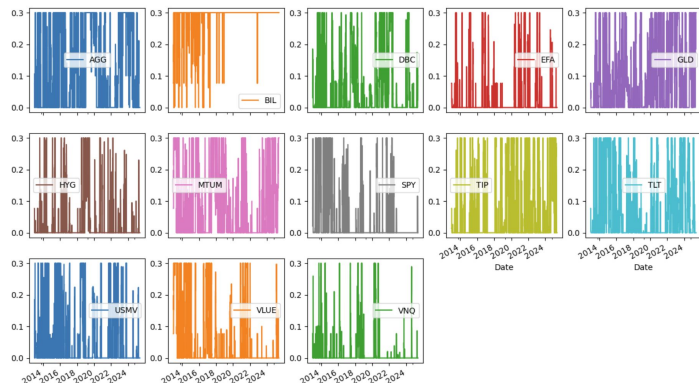
Exploring methods for Covariance Matrix: OAS

Alpha Overlay

Considerations

	# AGG	# BIL	# DBC	# EFA	# GLD	# HYG	# MTUM
count	3114.0	3114.0	3114.0	3114.0	3114.0	3114.0	3114.0
mean	0.12992090877152257	0.29043042074940867	0.06246598759096428	0.023763429226076088	0.0945283597510355	0.03184939456754767	0.08140214723599705
std	0.13472943549578673	0.04421866673929298	0.10119097250376564	0.06726357851094583	0.11771295840268804	0.07799444185767039	0.11138098183913228
min	0.0	0.0	0.0	0.0	0.0	0.0	0.0
25%	0.0	0.29999999996091165	0.0	0.0	0.0	0.0	0.0
50%	0.07692307692307694	0.3000000000541929	0.0	0.0	0.021758159860495258	0.0	0.0
75%	0.2999999996717096	0.3000000005590598	0.08976111316915458	0.0	0.18799744284701175	0.0	0.14712711328702172
max	0.30002184871667315	0.3000240706424491	0.3000196840286693	0.3000141198822103	0.3000059669989658	0.30002164772472456	0.3000199994739831

Alpha Model Weights using OAS Shrinkage Covariance Matrix



# SPY	# TIP	# TLT	# USMV	# VLU	# VNQ
3114.0	3114.0	3114.0	3114.0	3114.0	3114.0
0.033492839703834126	0.06506017886831936	0.06791073048034114	0.05441372140295624	0.05064693713279123	0.014114944519206061
0.07934407155070662	0.10615333400866513	0.10336500975889891	0.09973035099159704	0.0988160809369419	0.05036331767362976
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.099999999993166064	0.10500931141181999	0.06272195857081778	0.04288428977914906	0.0
0.3000005222519691	0.30002164762198	0.3000009139233718	0.3000002199009369	0.3000199994739831	0.30000038736516527

Overview

ETF Selection

Regime Model

Asset Returns Model

Asset Risks

Portfolio Construction

Attribution

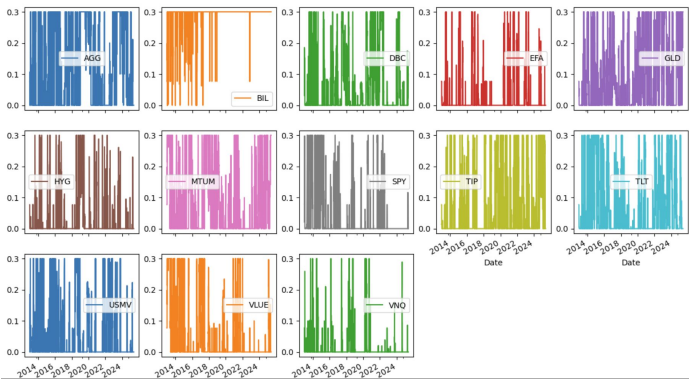
Modelling of Asset Risk: Covariance Matrix

Exploring methods for Covariance Matrix: Ledoit-Wolf

Considerations

	# AGG	# BIL	# DBC	# EFA	# GLD	# HYG	# MTUM
count	3114.0	3114.0	3114.0	3114.0	3114.0	3114.0	3114.0
mean	0.12992090877152257	0.29043042074940867	0.06246598759096428	0.023763429226076088	0.0945283597510355	0.03184939456754767	0.08140214723599705
std	0.13472943549578673	0.04421866673929298	0.10119097250376564	0.06726357851094583	0.11771295840268804	0.07799444185767039	0.11138098183913228
min	0.0	0.0	0.0	0.0	0.0	0.0	0.0
25%	0.0	0.29999999996091165	0.0	0.0	0.0	0.0	0.0
50%	0.07692307692307694	0.3000000000541929	0.0	0.0	0.021758159860495258	0.0	0.0
75%	0.2999999996717096	0.3000000005590598	0.08976111316915458	0.0	0.18799744284701175	0.0	0.14712711328702172
max	0.30002184871667315	0.3000240706424491	0.3000196840286693	0.3000141198822103	0.3000059669989658	0.30002164772472456	0.3000199994739831

Alpha Model Weights using Ledoit Wolf Shrinkage Covariance Matrix



# SPV	# TIP	# TLT	# USMV	# VLU	# VNQ
3114.0	3114.0	3114.0	3114.0	3114.0	3114.0
0.033492839703834126	0.06506017886831936	0.06791073048034114	0.05441372140295624	0.05064693713279123	0.014114944519206061
0.07934407155070662	0.10615333400866513	0.10336500975889891	0.09973035099159704	0.0988160809369419	0.05036331767362976
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.09999999993166064	0.10500931141181999	0.06272195857081778	0.04288428977914906	0.0
0.3000005222519691	0.30002164762198	0.3000009139233718	0.3000002199009369	0.3000199994739831	0.30000038736516527

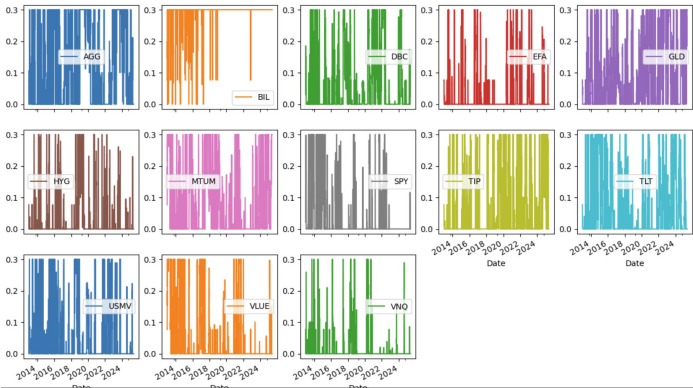
Modelling of Asset Risk: Covariance Matrix

Exploring methods for Covariance Matrix: Sample

Considerations

	# AGG	# BIL	# DBC	# EFA	# GLD	# HYG	# MTUM
count	3114.0	3114.0	3114.0	3114.0	3114.0	3114.0	3114.0
mean	0.12992090877152257	0.29043042074940867	0.06246598759096428	0.023763429226076088	0.0945283597510355	0.03184939456754767	0.08140214723599705
std	0.13472943549578673	0.04421866673929298	0.10119097250376564	0.06726357851094583	0.11771295840268804	0.07799444185767039	0.11138098183913228
min	0.0	0.0	0.0	0.0	0.0	0.0	0.0
25%	0.0	0.29999999996091165	0.0	0.0	0.0	0.0	0.0
50%	0.07692307692307694	0.3000000000541929	0.0	0.0	0.021758159860495258	0.0	0.0
75%	0.2999999996717096	0.3000000005590598	0.08976111316915458	0.0	0.18799744284701175	0.0	0.14712711328702172
max	0.30002184871667315	0.3000240706424491	0.3000196840286693	0.3000141198822103	0.3000059669989658	0.30002164772472456	0.3000199994739831

Linear Model Weights using Sample Shrinkage Covariance Matrix



# SPY	# TIP	# TLT	# USMV	# VLUE	# VNO
3114.0	3114.0	3114.0	3114.0	3114.0	3114.0
0.033492839703834126	0.06506017886831936	0.06791073048034114	0.05441372140295624	0.05064693713279123	0.014114944519206061
0.07934407155070662	0.10615333400866513	0.10336500975889891	0.09973035099159704	0.0988160809369419	0.05036331767362976
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0
0.0	0.09999999993166064	0.10500931141181999	0.06272195857081778	0.04288428977914906	0.0
0.3000005222519691	0.30002164762198	0.3000009139233718	0.3000002199009369	0.3000199994739831	0.3000038736516527

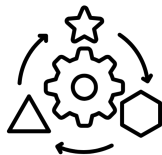
Portfolio Construction & Implementation

Objective, Model Selection

Objective: Construct an alpha overlay, well-diversified, portfolio, that is stable and automatically adapts to the changes in regime.

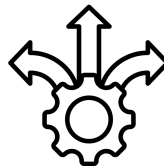
- Model of choice: Mean Variance Optimization (**MVO**)

Why MVO?



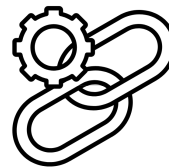
Adaptable

to regime changes and
new market conditions.
Express return views



Flexible

with constraints, easily
allowing us to incorporate
real-world constraints



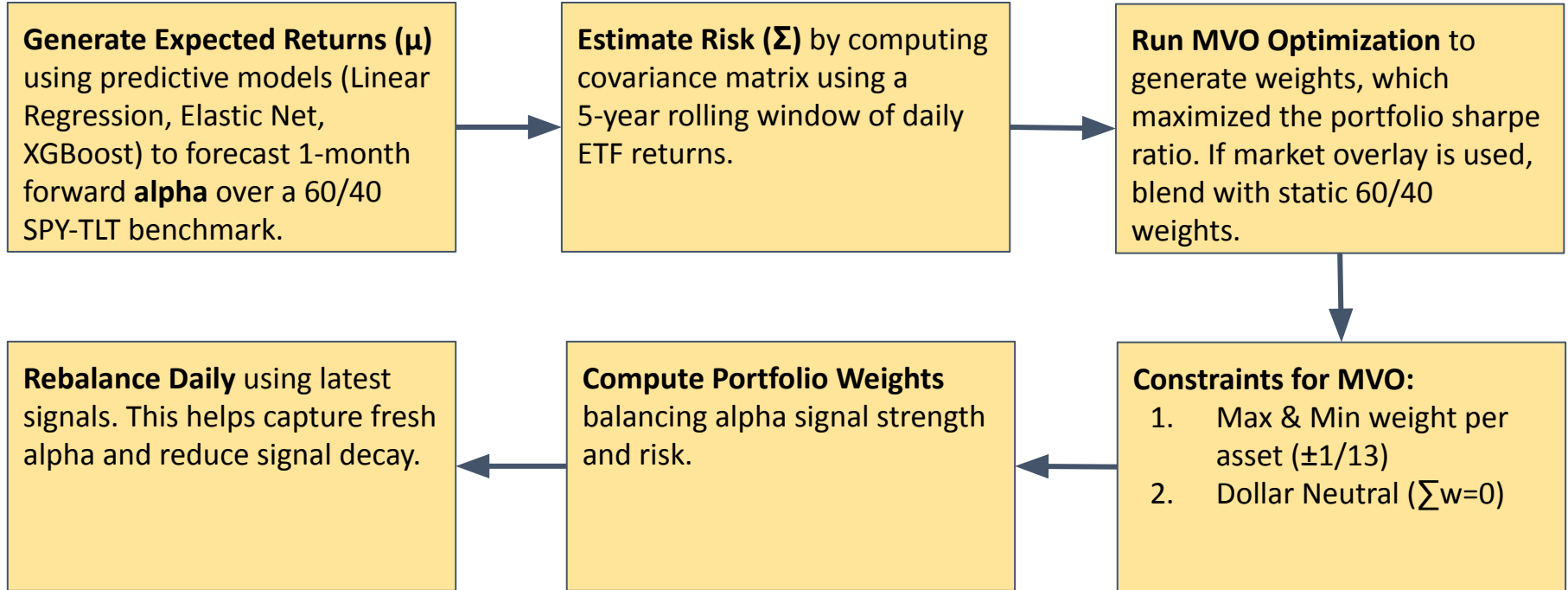
Robust

risk-management and
ensures diversified
portfolio

Portfolio Construction & Implementation

Methodology

Methodology:



Portfolio Construction & Implementation

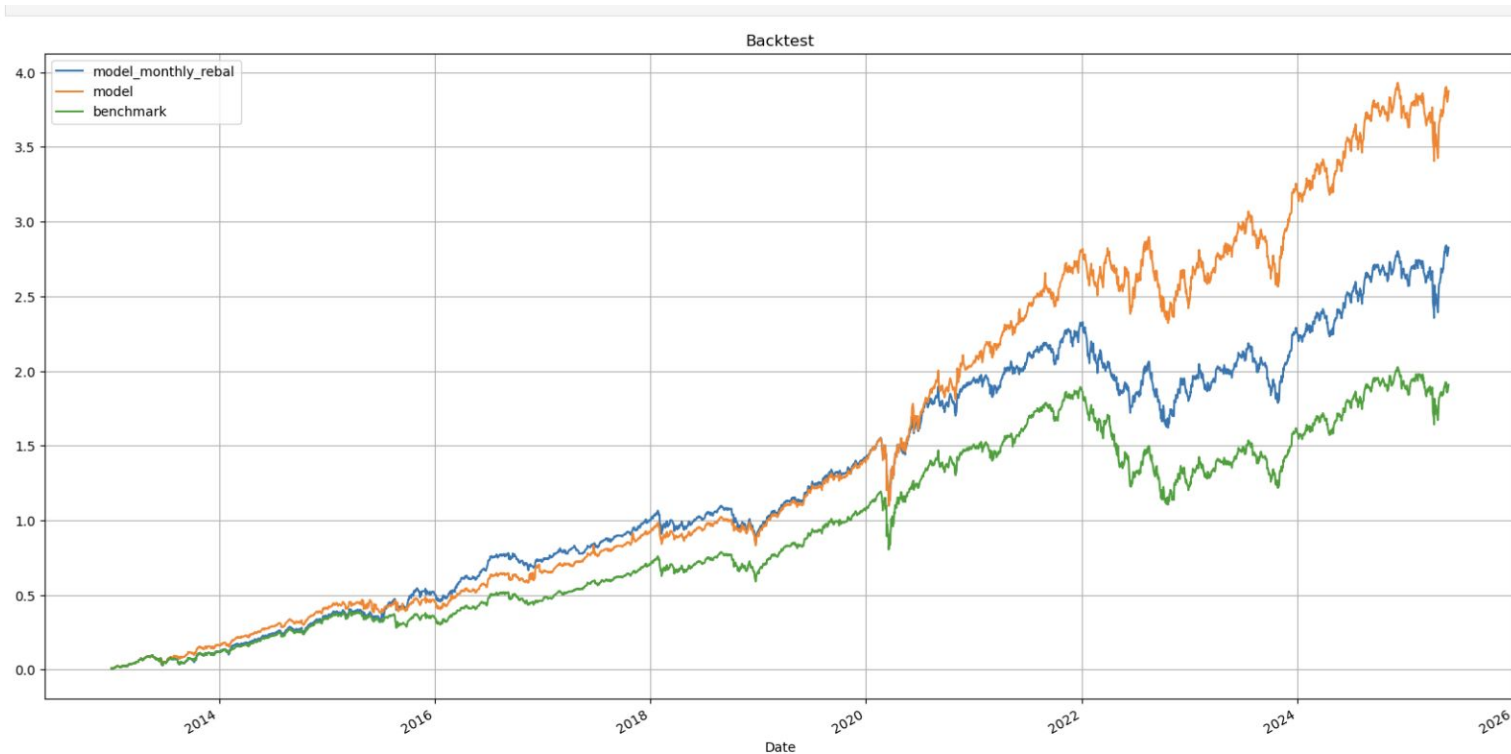
Transaction Cost Assumptions:

Ticker	Category	Typical Cost Range	Transaction Cost Estimated
SPY	Broad Market Index ETF	0.00% – 0.10%	5 bps (0.05%)
AGG	U.S. Bond ETF	0.05% – 0.15%	10 bps (0.10%)
GLD	Commodity ETF (Gold)	0.25% – 0.75%	40 bps (0.40%)
TLT	U.S. Bond ETF (long duration)	0.05% – 0.15%	10 bps (0.10%)
HYG	U.S. Bond ETF (high yield)	0.05% – 0.15%	12 bps (0.12%)
DBC	Commodity ETF (broad basket)	0.25% – 0.75%	50 bps (0.50%)
EFA	Intl Developed Market ETF	0.12% – 0.30%	20 bps (0.20%)
MTUM	Factor ETF (Momentum)	0.12% – 0.30%	20 bps (0.20%)
TIP	U.S. Bond ETF (inflation-linked)	0.05% – 0.15%	10 bps (0.10%)
BIL	U.S. Treasury (ultra-short)	0.00% – 0.10%	5 bps (0.05%)
USMV	Factor ETF (Min Volatility)	0.12% – 0.30%	20 bps (0.20%)
VLUE	Factor ETF (Value)	0.12% – 0.30%	20 bps (0.20%)
VNQ	Real Estate ETF (REITs)	0.12% – 0.50%	30 bps (0.30%)

Source: [Investopedia](<https://www.investopedia.com/ask/answers/071816/how-are-etf-fees-deducted.asp>)

Portfolio Performance

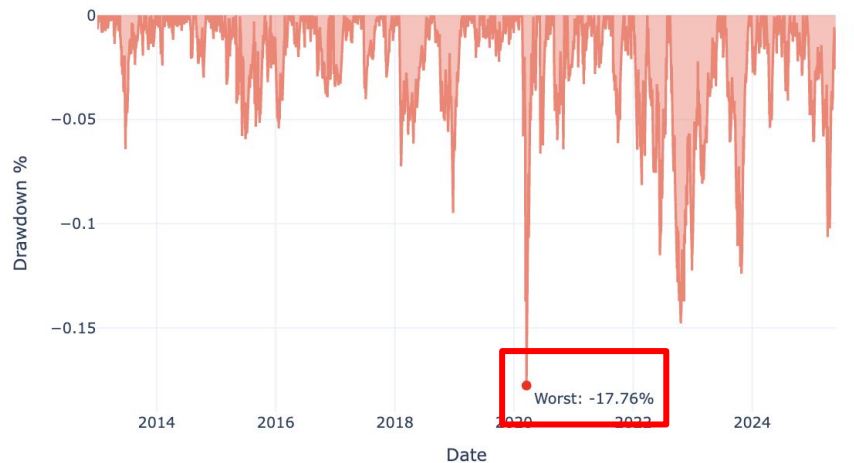
Backtest Results:



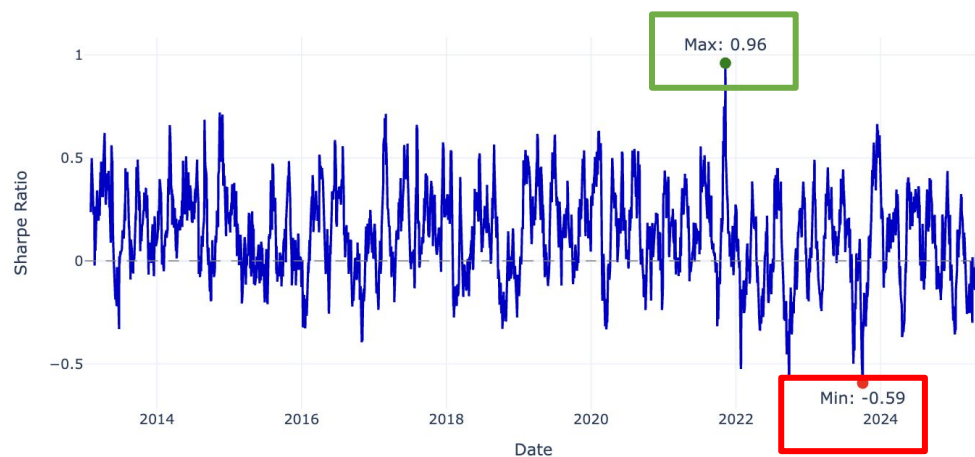
Portfolio Performance

Key Performance Metrics

Drawdown



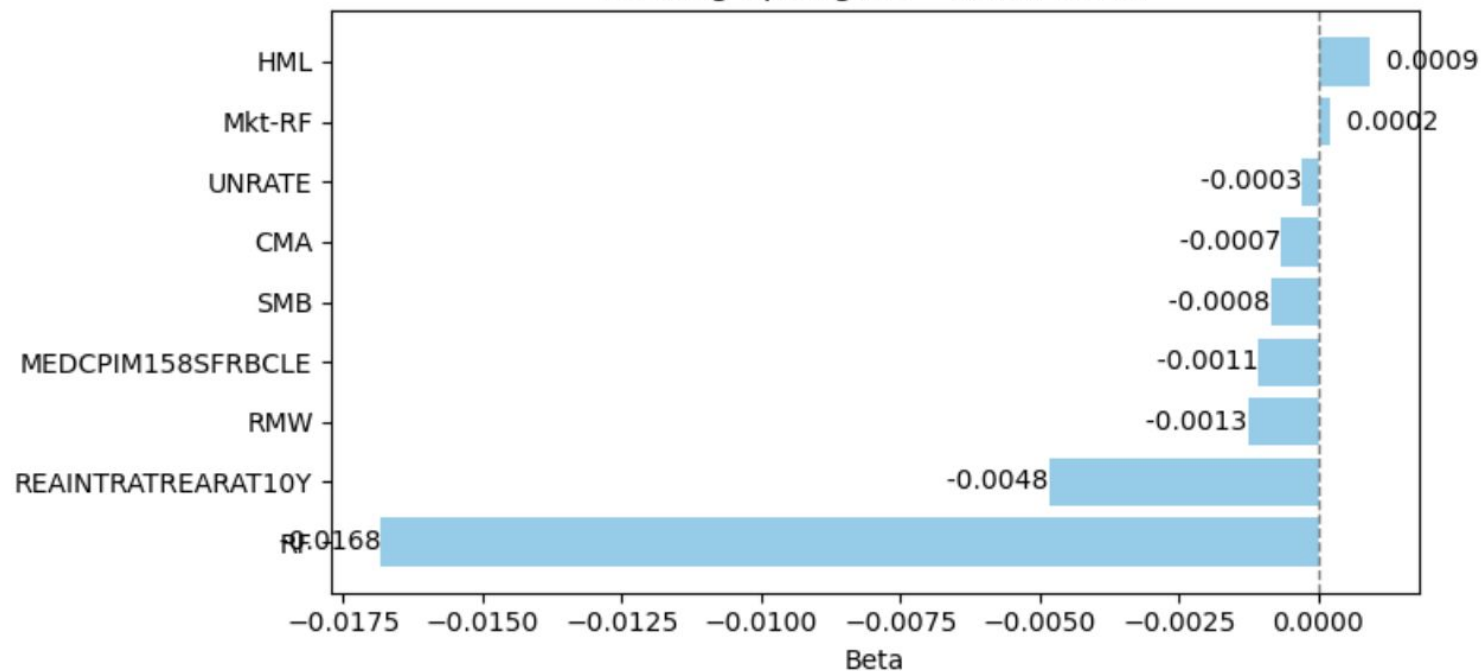
21 Day Rolling Sharpe Ratio



Portfolio Attribution - Heating Up

Key Performance Drivers

Heating Up Regime - Factor Betas



[Overview](#)

[ETF Selection](#)

[Regime Model](#)

[Asset Returns Model](#)

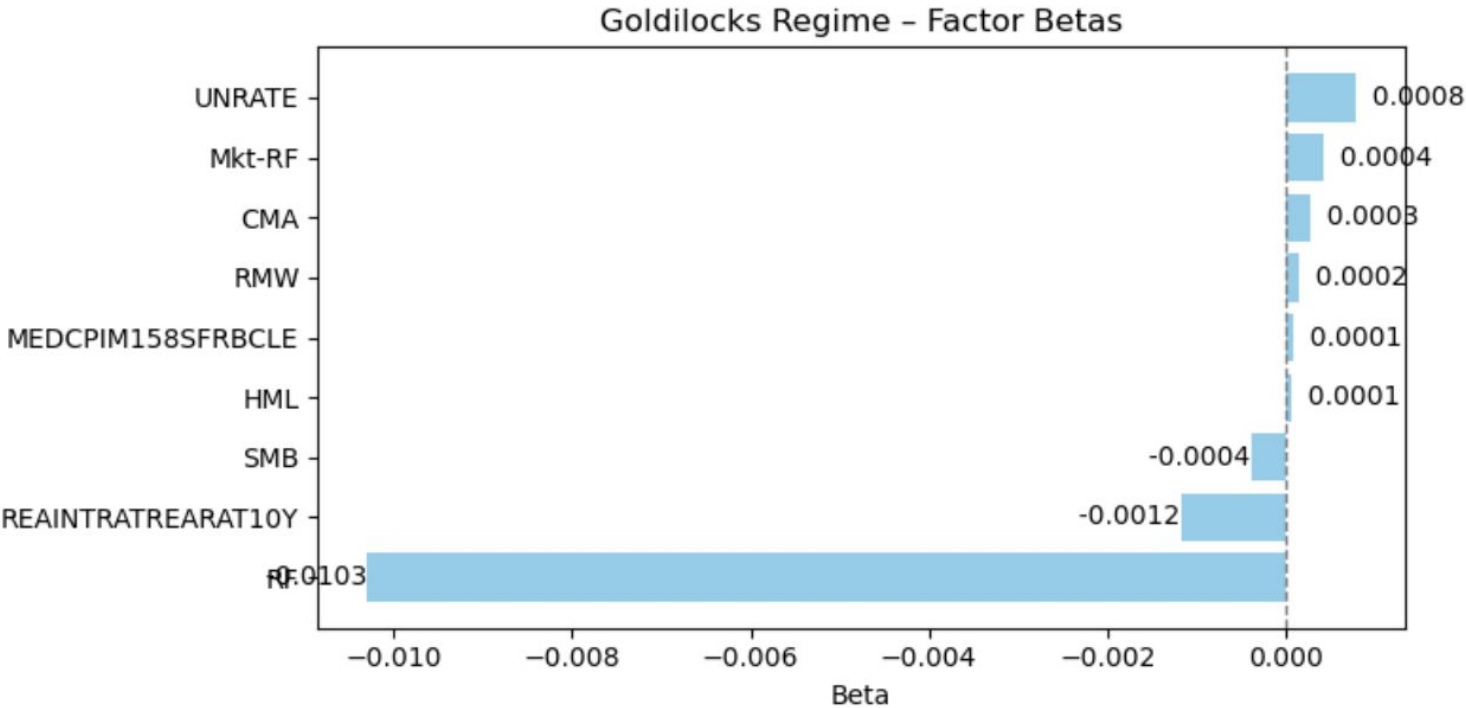
[Asset Risks](#)

[Portfolio Construction](#)

Attribution

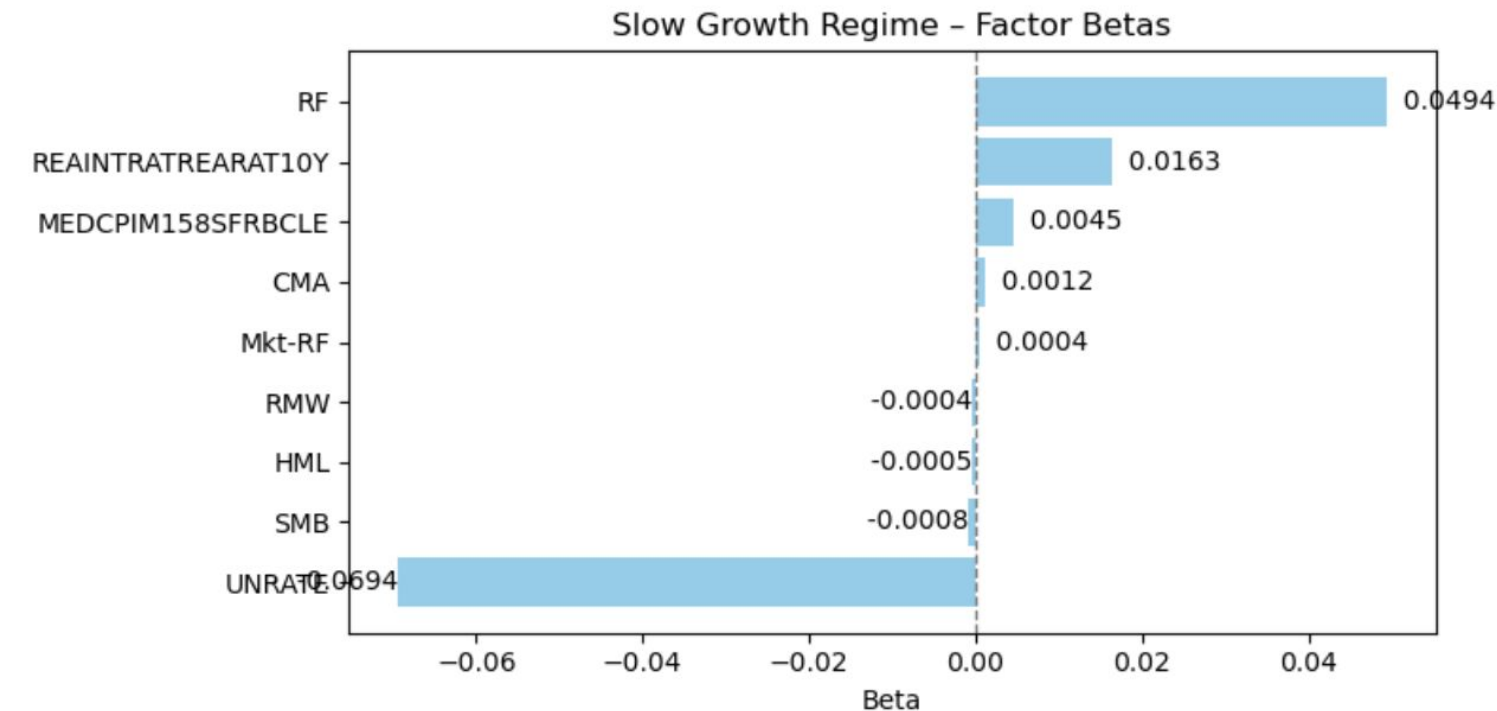
Portfolio Attribution - Goldilocks

Key Performance Drivers



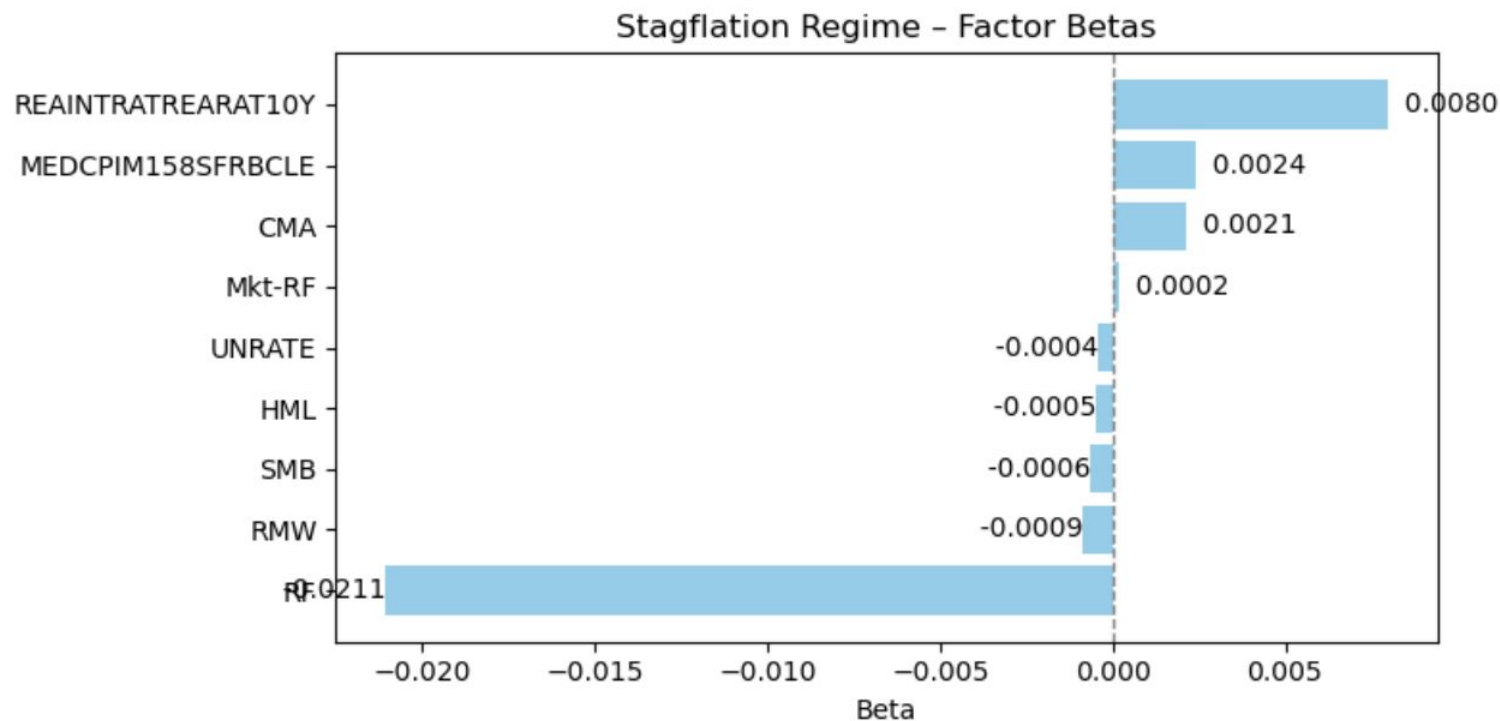
Portfolio Attribution - Slow Growth

Key Performance Drivers



Portfolio Attribution - Stagflation

Key Performance Drivers



[Overview](#)

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[Regime Model](#)

[Asset Returns Model](#)

[Asset Risks](#)

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Attribution

Future Model Improvements

What Could Be Done Differently

1. Exposure to growth and rates

- Scale RF carry exposure dynamically with yield-curve steepness (10Y-3M) to boost RF/AGG weight when the curve steepens.
- Implement an adaptive duration tilt by shifting a portion of IG bond weight into TLT when UNRATE rises MoM.
- Stress-test Fed-pivot scenarios (e.g. 50 bps cut) and establish stop-loss triggers to auto-reduce carry positions on sharp rate moves.

2. Equity and style tilts

- Optimize factor weights via a grid search for maximum Information Ratio across MKT, SMB, CMA signals.
- Scale SMB and CMA tilts inversely to their realized volatility to smooth style exposures in choppy equity markets.
- Introduce a quality overlay by constraining model to allocate to USMV or a QMJ basket whenever RMW β drops below -0.02%.

3. Sensitivity to inflation and quality

- Hedge inflation spikes by allocating 5-10% to TIP and DBC when CPI surprises exceed +50 bps.
- Activate a “quality floor” by reserving 5% for USMV whenever RMW β is negative to cushion high-vol environments.
- Incorporate 5Y breakeven-inflation as a gating filter to mute momentum signals when inflation expectations hit the 90th percentile.

4. Regime-driven risks

- Enforce a hard constraint of $\geq 25\%$ in USMV + AGG when Slow Growth probability $> 50\%$ to cap momentum at 50%.
- Define a 10% defensive bucket that rotates among BIL, USMV, and AGG based on the highest regime probability.
- Adjust rebalance freq to quarterly during rapid regime shifts to reduce turnover costs and whipsaw losses.

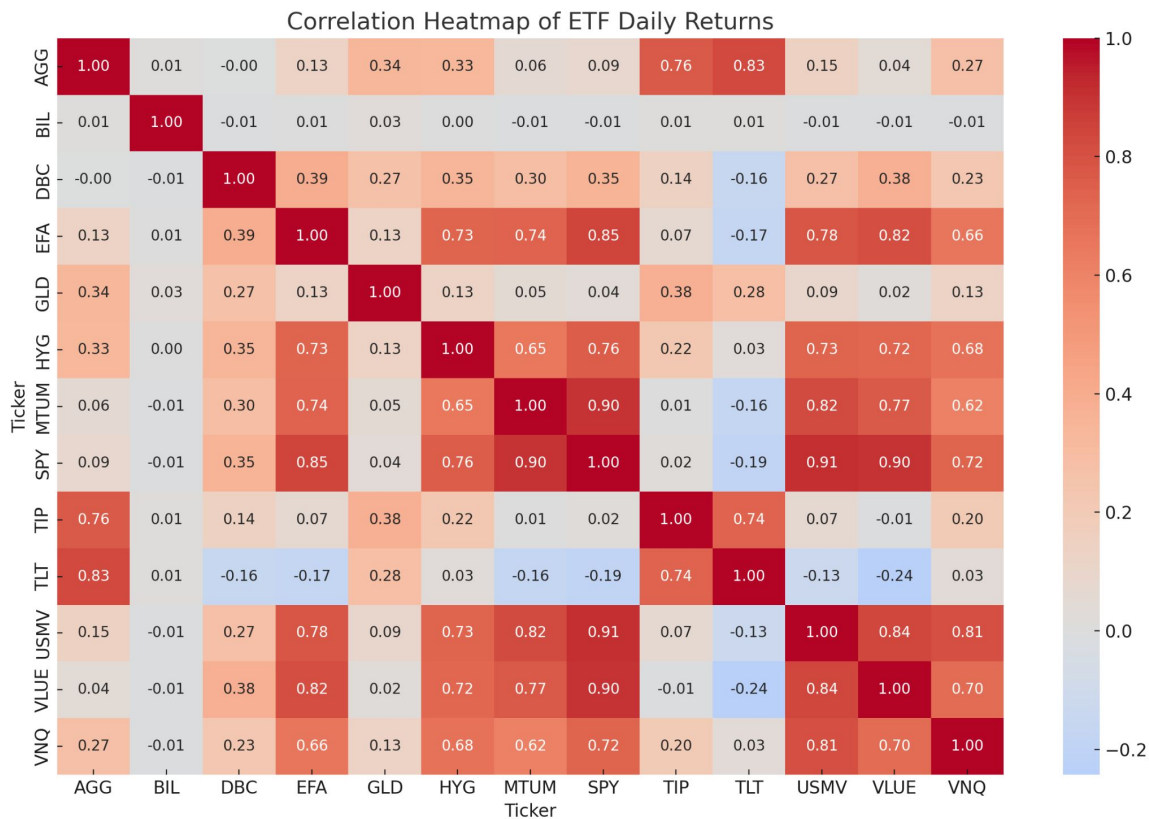
Conclusion and Discussion

How can we possibly improve?

- Optimize Rebalancing Frequency: Evaluate performance under bi-monthly or quarterly rebalancing to reduce turnover
- Use rolling smoothed portfolio weights to reduce turnover and lower transaction cost, due to lesser abrupt weight shifts.
- Regime benchmark smoothening: Due to naive assumption of MSCI fallback mechanism formula, benchmark regimes are too “noisy”. Applying quarterly adjustment will greatly reduce the noise to improve forecasting accuracy.

Appendix - ETF Selection

Correlation heatmap of ETF daily returns



Equities: SPY, MTUM, VLUE, USMV, EFA, VNQ

Fixed Income: AGG, HYG, TLT, TIP, BIL

Alternatives: GLD, DBC

Q: High correlation within the EQ sleeve?

A: Not a bug but a feature.

- The tickers differ on factor style
- High correlation to SPY → changes in MTUM/VLUE/USMV perf are less about broad market and more about style tilt
- This supports cleaner attribution later on.

